

Singularity Subtraction for the ERI Contraction in Periodic Coupled-Cluster Doubles

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Abstract

Xing and coworkers describe the effect (or lack thereof) of the Madelung correction applied to periodic coupled-cluster doubles (CCD). Here, we outline a procedure to address the correction of the ERI contraction with singularity subtraction, similar to exact exchange.

Chapter 1

Fitted Gaussian for each Madelung-invoking ERI Contraction

Disclaimer

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1. The introduction was mostly written by Codex, verified by the author.
2. The singularity subtraction strategy presented here was created by the author, but written and formatted by Codex.

1.1 Introduction

1.1.1 CCD Contraction Map

Coupled-cluster doubles (CCD) is a post-Hartree–Fock wave-function method in which the cluster operator contains only double excitations. Relative to a Hartree–Fock reference determinant $|\Phi\rangle$, the CCD ansatz is

$$|\Psi_{\text{CCD}}\rangle = e^{\hat{T}_2}|\Phi\rangle, \quad \hat{T}_2 = \frac{1}{N_k} \sum_{\substack{ijab \\ \mathbf{k}_i, \mathbf{k}_j, \mathbf{k}_a \in \mathcal{K}}} T_{ijab}(\mathbf{k}_i, \mathbf{k}_j, \mathbf{k}_a) a_{a\mathbf{k}_a}^\dagger a_{b\mathbf{k}_b}^\dagger a_{j\mathbf{k}_j} a_{i\mathbf{k}_i}, \quad (1.1)$$

where i, j and a, b label occupied and virtual orbitals, respectively, and $\mathbf{k}_b$ is fixed by crystal-momentum conservation, $\mathbf{k}_i + \mathbf{k}_j - \mathbf{k}_a - \mathbf{k}_b \in \mathcal{L}^*$.

For a finite uniform mesh $\mathcal{K}$ containing N_k points, collect the amplitudes into $T^{N_k} = \{t_{IJ}^{AB}\}$ using composite indices $I = (i, \mathbf{k}_i)$, $J = (j, \mathbf{k}_j)$, $A = (a, \mathbf{k}_a)$, and $B = (b, \mathbf{k}_b)$. The CCD equations are a nonlinear fixed-point problem,

$$T^{N_k} = \left(\varepsilon^{N_k}\right)^{-1} \mathcal{A}_{N_k}(T^{N_k}), \quad \varepsilon_{IJAB}^{N_k} = \varepsilon_{i\mathbf{k}_i}^{N_k} + \varepsilon_{j\mathbf{k}_j}^{N_k} - \varepsilon_{a\mathbf{k}_a}^{N_k} - \varepsilon_{b\mathbf{k}_b}^{N_k}, \quad (1.2)$$

where $\mathcal{A}_{N_k}$ is the ERI-contraction map. Writing $\mathcal{P}_{IJ}(X_{IJ}^{AB}) = X_{IJ}^{AB} + X_{JI}^{BA}$, the finite- k map used

in the paper is

$$\begin{aligned}
[\mathcal{A}_{N_k}(T)]_{IJ}^{AB} = & \langle AB|IJ \rangle + \mathcal{P}_{IJ} \left(\sum_C \kappa_C^A t_{IJ}^{CB} - \sum_K \kappa_I^K t_{KJ}^{AB} \right) \\
& + \frac{1}{N_k} \sum_{KL} \chi_{IJ}^{KL} t_{KL}^{AB} + \frac{1}{N_k} \sum_{CD} \chi_{CD}^{AB} t_{IJ}^{CD} \\
& + \mathcal{P}_{IJ} \left\{ \frac{1}{N_k} \sum_{KC} \left[(2\chi_{IC}^{AK} - \chi_{CI}^{AK}) t_{KJ}^{CB} - \chi_{IC}^{AK} t_{KJ}^{BC} - \chi_{CJ}^{AK} t_{KI}^{BC} \right] \right\}. \quad (1.3)
\end{aligned}$$

The intermediates in this contraction map are

$$\kappa_C^A = -\frac{1}{N_k^2} \sum_{KLD} (2\langle KL|CD \rangle - \langle KL|DC \rangle) t_{KL}^{AD}, \quad (\text{L1})$$

$$\kappa_I^K = \frac{1}{N_k^2} \sum_{LCD} (2\langle KL|CD \rangle - \langle KL|DC \rangle) t_{IL}^{CD}, \quad (\text{L2})$$

$$\chi_{IJ}^{KL} = \langle KL|IJ \rangle + \frac{1}{N_k} \sum_{CD} \langle KL|CD \rangle t_{IJ}^{CD}, \quad (\text{L3})$$

$$\chi_{CD}^{AB} = \langle AB|CD \rangle, \quad (\text{L4})$$

$$\chi_{IC}^{AK} = \langle AK|IC \rangle + \frac{1}{2N_k} \sum_{LD} \left[(2\langle LK|DC \rangle - \langle LK|CD \rangle) t_{IL}^{AD} - \langle LK|DC \rangle t_{IL}^{DA} \right], \quad (\text{L5})$$

$$\chi_{CI}^{AK} = \langle AK|CI \rangle - \frac{1}{2N_k} \sum_{LD} \langle LK|CD \rangle t_{IL}^{DA}. \quad (\text{L6})$$

All composite-index sums enforce the same crystal-momentum conservation as Eq. (1.1). Equations (1.2)–(L6) therefore give the finite- k contraction map and the amplitude equation needed for its solution.

1.1.2 Madelung-corrected terms

The Madelung correction is introduced by shifting the Ewald Coulomb kernel, which changes each normalized ERI according to

$$\begin{aligned}
\langle n_1 \mathbf{k}_1, n_2 \mathbf{k}_2 | n_3 \mathbf{k}_3, n_4 \mathbf{k}_4 \rangle_\xi = & \langle n_1 \mathbf{k}_1, n_2 \mathbf{k}_2 | n_3 \mathbf{k}_3, n_4 \mathbf{k}_4 \rangle \\
& - N_k \xi \delta_{n_1 n_3} \delta_{n_2 n_4} \delta_{\mathbf{k}_1 \mathbf{k}_3} \delta_{\mathbf{k}_2 \mathbf{k}_4}. \quad (1.4)
\end{aligned}$$

Here, ξ is the mesh-dependent Madelung constant. The correction is therefore present only when both orbital pairs are identical and the ERI has zero momentum transfer (with the momentum Kronecker deltas understood modulo the reciprocal lattice). Equivalently, it subtracts the leading Coulomb singularity at the point sampled by the finite $\mathbf{k}$ mesh. This is the singularity-subtraction interpretation of the Madelung correction: the finite-mesh quadrature is applied to the nonsingular remainder, while the corresponding singular contribution is restored analytically through ξ [1].

In an ERI contraction, the factor N_k in Eq. (1.4) cancels the $1/N_k$ quadrature weight. Thus, each triggered ERI contributes an explicit $\pm\xi$ times the amplitude multiplying it. To display the six terms that trigger the correction, let $K = (k, \mathbf{k}_k)$, $L = (l, \mathbf{k}_l)$, $C = (c, \mathbf{k}_c)$, and $D = (d, \mathbf{k}_d)$; the unshown momentum indices are fixed by crystal-momentum conservation. For fixed external indices

$(i, j, a, b, \mathbf{k}_i, \mathbf{k}_j, \mathbf{k}_a)$, the six corrected linear terms following Eq. (B10) are

$$\begin{aligned}
& \frac{1}{N_k} \sum_{\mathbf{k}_k \in \mathcal{K}} \sum_{kl} \langle k\mathbf{k}_k, l\mathbf{k}_l | i\mathbf{k}_i, j\mathbf{k}_j \rangle t_{klab}(\mathbf{k}_k, \mathbf{k}_l, \mathbf{k}_a) - \xi t_{ijab}(\mathbf{k}_i, \mathbf{k}_j, \mathbf{k}_a), \\
& \frac{1}{N_k} \sum_{\mathbf{k}_c \in \mathcal{K}} \sum_{cd} \langle a\mathbf{k}_a, b\mathbf{k}_b | c\mathbf{k}_c, d\mathbf{k}_d \rangle t_{ijcd}(\mathbf{k}_i, \mathbf{k}_j, \mathbf{k}_c) - \xi t_{ijab}(\mathbf{k}_i, \mathbf{k}_j, \mathbf{k}_a), \\
& - \frac{1}{N_k} \sum_{\mathbf{k}_k \in \mathcal{K}} \sum_{kc} \langle a\mathbf{k}_a, k\mathbf{k}_k | c\mathbf{k}_c, i\mathbf{k}_i \rangle t_{kjcb}(\mathbf{k}_k, \mathbf{k}_j, \mathbf{k}_c) + \xi t_{ijab}(\mathbf{k}_i, \mathbf{k}_j, \mathbf{k}_a), \\
& - \frac{1}{N_k} \sum_{\mathbf{k}_k \in \mathcal{K}} \sum_{kc} \langle b\mathbf{k}_b, k\mathbf{k}_k | c\mathbf{k}_c, j\mathbf{k}_j \rangle t_{kica}(\mathbf{k}_k, \mathbf{k}_i, \mathbf{k}_c) + \xi t_{ijab}(\mathbf{k}_i, \mathbf{k}_j, \mathbf{k}_a), \\
& - \frac{1}{N_k} \sum_{\mathbf{k}_k \in \mathcal{K}} \sum_{kc} \langle a\mathbf{k}_a, k\mathbf{k}_k | c\mathbf{k}_c, j\mathbf{k}_j \rangle t_{kibc}(\mathbf{k}_k, \mathbf{k}_i, \mathbf{k}_b) + \xi t_{ijab}(\mathbf{k}_i, \mathbf{k}_j, \mathbf{k}_a), \\
& - \frac{1}{N_k} \sum_{\mathbf{k}_k \in \mathcal{K}} \sum_{kc} \langle b\mathbf{k}_b, k\mathbf{k}_k | c\mathbf{k}_c, i\mathbf{k}_i \rangle t_{kjac}(\mathbf{k}_k, \mathbf{k}_j, \mathbf{k}_a) + \xi t_{ijab}(\mathbf{k}_i, \mathbf{k}_j, \mathbf{k}_a). \tag{1.5}
\end{aligned}$$

The two positive terms contribute $-2\xi T$, while the four negative terms contribute $+4\xi T$. Hence the complete ERI-contraction map is

$$\mathcal{A}_{N_k, \xi}(T) = \mathcal{A}_{N_k}(T) + 2\xi T. \tag{1.6}$$

1.2 Singularity Subtraction for the ERI Contraction

For each of the ERIS in (1.5), we can rewrite them in the usual reciprocal-space summation.

The transfer momentum is line-specific in the six expressions below. We use the following definitions, with the equivalent forms following from crystal-momentum conservation:

$$\begin{aligned}
\mathbf{q}_1 &= \mathbf{k}_i - \mathbf{k}_k = \mathbf{k}_l - \mathbf{k}_j, & \mathbf{q}_2 &= \mathbf{k}_c - \mathbf{k}_a = \mathbf{k}_b - \mathbf{k}_d, \\
\mathbf{q}_3 &= \mathbf{k}_c - \mathbf{k}_a = \mathbf{k}_k - \mathbf{k}_i, & \mathbf{q}_4 &= \mathbf{k}_c - \mathbf{k}_b = \mathbf{k}_k - \mathbf{k}_j, \\
\mathbf{q}_5 &= \mathbf{k}_c - \mathbf{k}_a = \mathbf{k}_k - \mathbf{k}_j, & \mathbf{q}_6 &= \mathbf{k}_c - \mathbf{k}_b = \mathbf{k}_k - \mathbf{k}_i.
\end{aligned}$$

$$[L_1]_{ijab, \mathbf{k}_i, \mathbf{k}_j, \mathbf{k}_a} = \frac{4\pi}{N_k |\Omega|} \sum_{\mathbf{q}_1 \in K_q} \sum'_{\mathbf{G} \in \mathbb{L}^*} \sum_{kl} \frac{\hat{\rho}_{k(\mathbf{k}_i - \mathbf{q}_1) i \mathbf{k}_i}(\mathbf{G}) \hat{\rho}_{l(\mathbf{k}_j + \mathbf{q}_1) j \mathbf{k}_j}(-\mathbf{G})}{|\mathbf{q}_1 + \mathbf{G}|^2} t_{kl}^{ab}(\mathbf{k}_i - \mathbf{q}_1, \mathbf{k}_j + \mathbf{q}_1, \mathbf{k}_a), \quad (1.7)$$

$$[L_2]_{ijab, \mathbf{k}_i, \mathbf{k}_j, \mathbf{k}_a} = \frac{4\pi}{N_k |\Omega|} \sum_{\mathbf{q}_2 \in K_q} \sum'_{\mathbf{G} \in \mathbb{L}^*} \sum_{cd} \frac{\hat{\rho}_{a \mathbf{k}_a c(\mathbf{k}_a + \mathbf{q}_2)}(\mathbf{G}) \hat{\rho}_{b \mathbf{k}_b d(\mathbf{k}_b - \mathbf{q}_2)}(-\mathbf{G})}{|\mathbf{q}_2 + \mathbf{G}|^2} t_{ij}^{cd}(\mathbf{k}_i, \mathbf{k}_j, \mathbf{k}_a + \mathbf{q}_2), \quad (1.8)$$

$$[L_3]_{ijab, \mathbf{k}_i, \mathbf{k}_j, \mathbf{k}_a} = -\frac{4\pi}{N_k |\Omega|} \sum_{\mathbf{q}_3 \in K_q} \sum'_{\mathbf{G} \in \mathbb{L}^*} \sum_{kc} \frac{\hat{\rho}_{a \mathbf{k}_a c(\mathbf{k}_a + \mathbf{q}_3)}(\mathbf{G}) \hat{\rho}_{k(\mathbf{k}_i + \mathbf{q}_3) i \mathbf{k}_i}(-\mathbf{G})}{|\mathbf{q}_3 + \mathbf{G}|^2} t_{kj}^{cb}(\mathbf{k}_i + \mathbf{q}_3, \mathbf{k}_j, \mathbf{k}_a + \mathbf{q}_3), \quad (1.9)$$

$$[L_4]_{ijab, \mathbf{k}_i, \mathbf{k}_j, \mathbf{k}_a} = -\frac{4\pi}{N_k |\Omega|} \sum_{\mathbf{q}_4 \in K_q} \sum'_{\mathbf{G} \in \mathbb{L}^*} \sum_{kc} \frac{\hat{\rho}_{b \mathbf{k}_b c(\mathbf{k}_b + \mathbf{q}_4)}(\mathbf{G}) \hat{\rho}_{k(\mathbf{k}_j + \mathbf{q}_4) j \mathbf{k}_j}(-\mathbf{G})}{|\mathbf{q}_4 + \mathbf{G}|^2} t_{ki}^{ca}(\mathbf{k}_j + \mathbf{q}_4, \mathbf{k}_i, \mathbf{k}_b + \mathbf{q}_4), \quad (1.10)$$

$$[L_5]_{ijab, \mathbf{k}_i, \mathbf{k}_j, \mathbf{k}_a} = -\frac{4\pi}{N_k |\Omega|} \sum_{\mathbf{q}_5 \in K_q} \sum'_{\mathbf{G} \in \mathbb{L}^*} \sum_{kc} \frac{\hat{\rho}_{a \mathbf{k}_a c(\mathbf{k}_a + \mathbf{q}_5)}(\mathbf{G}) \hat{\rho}_{k(\mathbf{k}_j + \mathbf{q}_5) j \mathbf{k}_j}(-\mathbf{G})}{|\mathbf{q}_5 + \mathbf{G}|^2} t_{ki}^{bc}(\mathbf{k}_j + \mathbf{q}_5, \mathbf{k}_i, \mathbf{k}_b), \quad (1.11)$$

$$[L_6]_{ijab, \mathbf{k}_i, \mathbf{k}_j, \mathbf{k}_a} = -\frac{4\pi}{N_k |\Omega|} \sum_{\mathbf{q}_6 \in K_q} \sum'_{\mathbf{G} \in \mathbb{L}^*} \sum_{kc} \frac{\hat{\rho}_{b \mathbf{k}_b c(\mathbf{k}_b + \mathbf{q}_6)}(\mathbf{G}) \hat{\rho}_{k(\mathbf{k}_i + \mathbf{q}_6) i \mathbf{k}_i}(-\mathbf{G})}{|\mathbf{q}_6 + \mathbf{G}|^2} t_{kj}^{ac}(\mathbf{k}_i + \mathbf{q}_6, \mathbf{k}_j, \mathbf{k}_a). \quad (1.12)$$

1.2.1 Singularity subtraction for L_1

Following the construction in Eqs. (13)–(14) of the ExxSS procedure, define the structure factor for L_1 as the complete numerator that multiplies the Coulomb kernel:

$$[S_1(\mathbf{q}_1 + \mathbf{G})]_{ijab, \mathbf{k}_i, \mathbf{k}_j, \mathbf{k}_a} = \sum_{kl} \hat{\rho}_{k(\mathbf{k}_i - \mathbf{q}_1) i \mathbf{k}_i}(\mathbf{G}) \hat{\rho}_{l(\mathbf{k}_j + \mathbf{q}_1) j \mathbf{k}_j}(-\mathbf{G}) \times t_{kl}^{ab}(\mathbf{k}_i - \mathbf{q}_1, \mathbf{k}_j + \mathbf{q}_1, \mathbf{k}_a). \quad (1.13)$$

At the singular point, orbital orthonormality gives

$$[S_1(\mathbf{0})]_{ijab, \mathbf{k}_i, \mathbf{k}_j, \mathbf{k}_a} = t_{ij}^{ab}(\mathbf{k}_i, \mathbf{k}_j, \mathbf{k}_a), \quad (1.14)$$

Thus, for an auxiliary function $h_1(\mathbf{Q})$ satisfying $h_1(\mathbf{0}) = 1$, the singularity-subtracted form is

$$\begin{aligned} [L_1^{\text{SS}}[h_1]]_{ijab, \mathbf{k}_i, \mathbf{k}_j, \mathbf{k}_a} &= \frac{4\pi}{N_k |\Omega|} \sum_{\mathbf{q}_1 \in K_q} \sum'_{\mathbf{G} \in \mathbb{L}^*} \frac{[S_1(\mathbf{q}_1 + \mathbf{G})]_{ijab, \mathbf{k}_i, \mathbf{k}_j, \mathbf{k}_a} - t_{ij}^{ab}(\mathbf{k}_i, \mathbf{k}_j, \mathbf{k}_a) h_1(\mathbf{q}_1 + \mathbf{G})}{|\mathbf{q}_1 + \mathbf{G}|^2} \\ &\quad + \frac{1}{(2\pi)^3} \int_{\mathbb{R}^3} d\mathbf{Q} \frac{4\pi t_{ij}^{ab}(\mathbf{k}_i, \mathbf{k}_j, \mathbf{k}_a) h_1(\mathbf{Q})}{|\mathbf{Q}|^2}, \end{aligned} \quad (1.15)$$

1.2.2 Alternate definition of the L_1 structure factor

Alternatively, (1) fix $(k, l) = (i, j)$ and (2) set $\mathbf{q}_1 = \mathbf{0}$ only in the amplitude:

$$\left[\tilde{\mathcal{S}}_1(\mathbf{q}_1 + \mathbf{G}) \right]_{ijab, \mathbf{k}_i \mathbf{k}_j \mathbf{k}_a} = \hat{\rho}_{i(\mathbf{k}_i - \mathbf{q}_1) i \mathbf{k}_i}(\mathbf{G}) \hat{\rho}_{j(\mathbf{k}_j + \mathbf{q}_1) j \mathbf{k}_j}(-\mathbf{G}) t_{ij}^{ab}(\mathbf{k}_i, \mathbf{k}_j, \mathbf{k}_a). \quad (1.16)$$

This definition is more practically motivated. In the calculation of this term, we are primarily interested in the terms that have nonzero contributions to $\left[\tilde{\mathcal{S}}(\mathbf{0}) \right]_{ijab, \mathbf{k}_i \mathbf{k}_j \mathbf{k}_a}$, therefore addressing the leading order $\mathcal{O}(N_{\mathbf{k}}^{-1/3})$ error. This means that for each term L_n , we only consider the orbital indices that match, and we neglect the change in t_{ij}^{ab} as a function of $\mathbf{q}_n$. This prevents us from having to store the t amplitudes in the event of memory constraints or the development of an integral-direct routine for CCSD. This also reduce CPU time scaling by O^2 , V^2 , or OV depending on the term since we only need to consider the orbital indices that match.

The expected downside is that the Taylor expansion around $\left[\tilde{\mathcal{S}}_n(\mathbf{0}) \right]_{ijab, \mathbf{k}_i \mathbf{k}_j \mathbf{k}_a}$ beyond the constant is not as accurately captured. Constraint (2) may be relaxed if the t amplitudes are stored, and constraint (1) may be relaxed if the runtime cost of the additional orbital indices is not prohibitive.

1.2.3 Definition of the L_1 correction

Define the auxiliary integrand

$$Q_1(\mathbf{Q}) = \frac{4\pi h_1(\mathbf{Q})}{|\mathbf{Q}|^2}. \quad (1.17)$$

The constant added to the uncorrected L_1 term is the negative finite-mesh quadrature of Q_1 plus its true integral over $\mathbb{R}^3$:

$$[\xi_1[h_1]]_{ijab, \mathbf{k}_i \mathbf{k}_j \mathbf{k}_a} = -\frac{1}{N_k |\Omega|} \sum_{\mathbf{q}_1 \in K_q} \sum'_{\mathbf{G} \in \mathbb{L}^*} Q_1(\mathbf{q}_1 + \mathbf{G}) + \frac{1}{(2\pi)^3} \int_{\mathbb{R}^3} d\mathbf{Q} Q_1(\mathbf{Q}). \quad (1.18)$$

Consequently, Eq. (1.15) can be written compactly as

$$\left[L_1^{\text{SS}}[h_1] \right]_{ijab, \mathbf{k}_i, \mathbf{k}_j, \mathbf{k}_a} = [L_1]_{ijab, \mathbf{k}_i, \mathbf{k}_j, \mathbf{k}_a} + [\xi_1[h_1]]_{ijab, \mathbf{k}_i \mathbf{k}_j \mathbf{k}_a} t_{ij}^{ab}(\mathbf{k}_i, \mathbf{k}_j, \mathbf{k}_a). \quad (1.19)$$

The decay function h_1 can be chosen based on benchmarking, but for now we will just use a Gaussian of with σ . If $\tilde{\mathcal{S}}_n$ is computed with constraint (2), then all $[\xi_1[h_1]]_{ijab, \mathbf{k}_i \mathbf{k}_j \mathbf{k}_a}$ can actually be precomputed before the CCD iterations begin. Specifically, for each $ijab, \mathbf{k}_i, \mathbf{k}_j, \mathbf{k}_a$ and each term 1-6, compute

$$\frac{\left[\tilde{\mathcal{S}}_1(\mathbf{q}_1 + \mathbf{G}) \right]_{ijab, \mathbf{k}_i \mathbf{k}_j \mathbf{k}_a}}{t_{ij}^{ab}(\mathbf{k}_i, \mathbf{k}_j, \mathbf{k}_a)} \quad (1.20)$$

which should be normalized to 1 at $\mathbf{q}_1 + \mathbf{G} = \mathbf{0}$. Then fit h_1 to this, which should set $[\xi_1[h_1]]_{ijab, \mathbf{k}_i \mathbf{k}_j \mathbf{k}_a}$ and is simply reused and scaled by the t amplitudes at each iteration.

In the case where constraint (2) is not followed, we would need to do a fitting at each CCD iteration, which means $[\xi_1[h_1]]_{ijab, \mathbf{k}_i \mathbf{k}_j \mathbf{k}_a}$ would change slightly and cannot just be precomputed and reused.

The structure factors for L_2 through L_6 , along with their practical approximations, are collected in Appendix 1.A.

1.3 Caveats and Future Work

The above procedure mainly focuses on the ERI contractions. We have assumed that the orbital energies themselves are fully accurate. In practice, however, the orbital energies are also going to be shifted by the Madelung correction. We will need to see the effect of using either the Madelung-corrected orbital energies or ExxSS-corrected orbital energies.

Implementation will be on PySCF and will be tested on simple crystals at the cc-pVDZ level. Preliminary calculations suggest that up to 4x4x4 is accessible for diamond and 2x2x2 is accessible for ammonia.

1.A Structure factors for L_2 – L_6

For subsequent extension of the singularity-subtraction procedure, the structure factors associated with L_2 through L_6 are

$$[\mathcal{S}_2(\mathbf{q}_2 + \mathbf{G})]_{ijab, \mathbf{k}_i \mathbf{k}_j \mathbf{k}_a} = \sum_{cd} \hat{\rho}_{a\mathbf{k}_a c}(\mathbf{k}_a + \mathbf{q}_2)(\mathbf{G}) \hat{\rho}_{b\mathbf{k}_b d}(\mathbf{k}_b - \mathbf{q}_2)(-\mathbf{G}) t_{ij}^{cd}(\mathbf{k}_i, \mathbf{k}_j, \mathbf{k}_a + \mathbf{q}_2), \quad (1.21)$$

$$[\mathcal{S}_3(\mathbf{q}_3 + \mathbf{G})]_{ijab, \mathbf{k}_i \mathbf{k}_j \mathbf{k}_a} = \sum_{kc} \hat{\rho}_{a\mathbf{k}_a c}(\mathbf{k}_a + \mathbf{q}_3)(\mathbf{G}) \hat{\rho}_{k(\mathbf{k}_i + \mathbf{q}_3) i \mathbf{k}_i}(-\mathbf{G}) t_{kj}^{cb}(\mathbf{k}_i + \mathbf{q}_3, \mathbf{k}_j, \mathbf{k}_a + \mathbf{q}_3), \quad (1.22)$$

$$[\mathcal{S}_4(\mathbf{q}_4 + \mathbf{G})]_{ijab, \mathbf{k}_i \mathbf{k}_j \mathbf{k}_a} = \sum_{kc} \hat{\rho}_{b\mathbf{k}_b c}(\mathbf{k}_b + \mathbf{q}_4)(\mathbf{G}) \hat{\rho}_{k(\mathbf{k}_j + \mathbf{q}_4) j \mathbf{k}_j}(-\mathbf{G}) t_{ki}^{ca}(\mathbf{k}_j + \mathbf{q}_4, \mathbf{k}_i, \mathbf{k}_b + \mathbf{q}_4), \quad (1.23)$$

$$[\mathcal{S}_5(\mathbf{q}_5 + \mathbf{G})]_{ijab, \mathbf{k}_i \mathbf{k}_j \mathbf{k}_a} = \sum_{kc} \hat{\rho}_{a\mathbf{k}_a c}(\mathbf{k}_a + \mathbf{q}_5)(\mathbf{G}) \hat{\rho}_{k(\mathbf{k}_j + \mathbf{q}_5) j \mathbf{k}_j}(-\mathbf{G}) t_{ki}^{bc}(\mathbf{k}_j + \mathbf{q}_5, \mathbf{k}_i, \mathbf{k}_b). \quad (1.24)$$

$$[\mathcal{S}_6(\mathbf{q}_6 + \mathbf{G})]_{ijab, \mathbf{k}_i \mathbf{k}_j \mathbf{k}_a} = \sum_{kc} \hat{\rho}_{b\mathbf{k}_b c}(\mathbf{k}_b + \mathbf{q}_6)(\mathbf{G}) \hat{\rho}_{k(\mathbf{k}_i + \mathbf{q}_6) i \mathbf{k}_i}(-\mathbf{G}) t_{kj}^{ac}(\mathbf{k}_i + \mathbf{q}_6, \mathbf{k}_j, \mathbf{k}_a). \quad (1.25)$$

The practical alternatives follow the same two restrictions used in Eq. (1.16): retain only the orbital indices that contribute at the singular point, and evaluate the amplitude at $\mathbf{q}_n = \mathbf{0}$. They are

$$[\tilde{\mathcal{S}}_2(\mathbf{q}_2 + \mathbf{G})]_{ijab, \mathbf{k}_i \mathbf{k}_j \mathbf{k}_a} = \hat{\rho}_{a\mathbf{k}_a a}(\mathbf{k}_a + \mathbf{q}_2)(\mathbf{G}) \hat{\rho}_{b\mathbf{k}_b b}(\mathbf{k}_b - \mathbf{q}_2)(-\mathbf{G}) t_{ij}^{ab}(\mathbf{k}_i, \mathbf{k}_j, \mathbf{k}_a), \quad (1.26)$$

$$[\tilde{\mathcal{S}}_3(\mathbf{q}_3 + \mathbf{G})]_{ijab, \mathbf{k}_i \mathbf{k}_j \mathbf{k}_a} = \hat{\rho}_{a\mathbf{k}_a a}(\mathbf{k}_a + \mathbf{q}_3)(\mathbf{G}) \hat{\rho}_{i(\mathbf{k}_i + \mathbf{q}_3) i \mathbf{k}_i}(-\mathbf{G}) t_{ij}^{ab}(\mathbf{k}_i, \mathbf{k}_j, \mathbf{k}_a), \quad (1.27)$$

$$[\tilde{\mathcal{S}}_4(\mathbf{q}_4 + \mathbf{G})]_{ijab, \mathbf{k}_i \mathbf{k}_j \mathbf{k}_a} = \hat{\rho}_{b\mathbf{k}_b b}(\mathbf{k}_b + \mathbf{q}_4)(\mathbf{G}) \hat{\rho}_{j(\mathbf{k}_j + \mathbf{q}_4) j \mathbf{k}_j}(-\mathbf{G}) t_{ji}^{ba}(\mathbf{k}_j, \mathbf{k}_i, \mathbf{k}_b), \quad (1.28)$$

$$[\tilde{\mathcal{S}}_5(\mathbf{q}_5 + \mathbf{G})]_{ijab, \mathbf{k}_i \mathbf{k}_j \mathbf{k}_a} = \hat{\rho}_{a\mathbf{k}_a a}(\mathbf{k}_a + \mathbf{q}_5)(\mathbf{G}) \hat{\rho}_{j(\mathbf{k}_j + \mathbf{q}_5) j \mathbf{k}_j}(-\mathbf{G}) t_{ji}^{ba}(\mathbf{k}_j, \mathbf{k}_i, \mathbf{k}_b), \quad (1.29)$$

$$[\tilde{\mathcal{S}}_6(\mathbf{q}_6 + \mathbf{G})]_{ijab, \mathbf{k}_i \mathbf{k}_j \mathbf{k}_a} = \hat{\rho}_{b\mathbf{k}_b b}(\mathbf{k}_b + \mathbf{q}_6)(\mathbf{G}) \hat{\rho}_{i(\mathbf{k}_i + \mathbf{q}_6) i \mathbf{k}_i}(-\mathbf{G}) t_{ij}^{ab}(\mathbf{k}_i, \mathbf{k}_j, \mathbf{k}_a). \quad (1.30)$$

Each expression satisfies $[\tilde{\mathcal{S}}_n(\mathbf{0})]_{ijab, \mathbf{k}_i \mathbf{k}_j \mathbf{k}_a} = t_{ij}^{ab}$ after applying the CCD pair symmetry $t_{ji}^{ba} = t_{ij}^{ab}$.

The corresponding singularity-subtraction corrections are obtained by replacing $\mathcal{S}_1$, t_{ij}^{ab} , and h_1 in Eq. (1.15) with the channel-specific $\tilde{\mathcal{S}}_n$, $\tilde{\mathcal{S}}_n(\mathbf{0})$, and h_n , respectively, while retaining the sign of L_n .

Chapter 2

Resolving imaginary components of the structure factor

Debugging the implementation in the prior chapter reveals that a real auxiliary function is being used to fit an imaginary function. This is because we are attempting to fit the structure factors evaluated at each $\mathbf{k}_i, \mathbf{k}_j, \mathbf{k}_a$ point rather than summing over the $\mathbf{k}$ -grids. Therefore, time-reversal symmetry is not used to cancel out imaginary components.

Here, we define structure factors for each of the L_i terms that contract over the appropriate k -points. Starting with L_1 we have that

$$\mathcal{S}_1(\mathbf{q}_1 + \mathbf{G}) = \sum_{ijkl} \sum_{\mathbf{k}_i, \mathbf{k}_j, \mathbf{k}_a} \hat{\rho}_{k(\mathbf{k}_i - \mathbf{q}_1)i\mathbf{k}_i}(\mathbf{G}) \hat{\rho}_{l(\mathbf{k}_j + \mathbf{q}_1)j\mathbf{k}_j}(-\mathbf{G}) t_{kl}^{ab}(\mathbf{k}_i - \mathbf{q}_1, \mathbf{k}_j + \mathbf{q}_1, \mathbf{k}_a). \quad (2.1)$$

For the corresponding practical approximation, we set $\mathbf{q}_1 = \mathbf{0}$ in the amplitude and retain only the terms with $(k, l) = (i, j)$:

$$\begin{aligned} \tilde{\mathcal{S}}_1(\mathbf{q}_1 + \mathbf{G}) &= \sum_{ijab} \sum_{\mathbf{k}_i, \mathbf{k}_j, \mathbf{k}_a} \hat{\rho}_{i(\mathbf{k}_i - \mathbf{q}_1)i\mathbf{k}_i}(\mathbf{G}) \hat{\rho}_{j(\mathbf{k}_j + \mathbf{q}_1)j\mathbf{k}_j}(-\mathbf{G}) \\ &\quad \times t_{ij}^{ab}(\mathbf{k}_i, \mathbf{k}_j, \mathbf{k}_a). \end{aligned} \quad (2.2)$$

The corresponding summed structure factors for L_2 through L_6 are

$$\begin{aligned} \mathcal{S}_2(\mathbf{q}_2 + \mathbf{G}) &= \sum_{ijkl} \sum_{\mathbf{k}_i, \mathbf{k}_j, \mathbf{k}_a} \hat{\rho}_{a\mathbf{k}_a c}(\mathbf{k}_a + \mathbf{q}_2)(\mathbf{G}) \hat{\rho}_{b\mathbf{k}_b d}(\mathbf{k}_b - \mathbf{q}_2)(-\mathbf{G}) \\ &\quad \times t_{ij}^{cd}(\mathbf{k}_i, \mathbf{k}_j, \mathbf{k}_a + \mathbf{q}_2), \end{aligned} \quad (2.3)$$

$$\begin{aligned} \mathcal{S}_3(\mathbf{q}_3 + \mathbf{G}) &= \sum_{ijkl} \sum_{\mathbf{k}_i, \mathbf{k}_j, \mathbf{k}_a} \hat{\rho}_{a\mathbf{k}_a c}(\mathbf{k}_a + \mathbf{q}_3)(\mathbf{G}) \hat{\rho}_{k(\mathbf{k}_i + \mathbf{q}_3)i\mathbf{k}_i}(-\mathbf{G}) \\ &\quad \times t_{kj}^{cb}(\mathbf{k}_i + \mathbf{q}_3, \mathbf{k}_j, \mathbf{k}_a + \mathbf{q}_3), \end{aligned} \quad (2.4)$$

$$\begin{aligned} \mathcal{S}_4(\mathbf{q}_4 + \mathbf{G}) &= \sum_{ijkl} \sum_{\mathbf{k}_i, \mathbf{k}_j, \mathbf{k}_a} \hat{\rho}_{b\mathbf{k}_b c}(\mathbf{k}_b + \mathbf{q}_4)(\mathbf{G}) \hat{\rho}_{k(\mathbf{k}_j + \mathbf{q}_4)j\mathbf{k}_j}(-\mathbf{G}) \\ &\quad \times t_{ki}^{ca}(\mathbf{k}_j + \mathbf{q}_4, \mathbf{k}_i, \mathbf{k}_b + \mathbf{q}_4), \end{aligned} \quad (2.5)$$

$$\begin{aligned} \mathcal{S}_5(\mathbf{q}_5 + \mathbf{G}) &= \sum_{ijkl} \sum_{\mathbf{k}_i, \mathbf{k}_j, \mathbf{k}_a} \hat{\rho}_{a\mathbf{k}_a c}(\mathbf{k}_a + \mathbf{q}_5)(\mathbf{G}) \hat{\rho}_{k(\mathbf{k}_j + \mathbf{q}_5)j\mathbf{k}_j}(-\mathbf{G}) \\ &\quad \times t_{ki}^{bc}(\mathbf{k}_j + \mathbf{q}_5, \mathbf{k}_i, \mathbf{k}_b), \end{aligned} \quad (2.6)$$

$$\begin{aligned} \mathcal{S}_6(\mathbf{q}_6 + \mathbf{G}) &= \sum_{ijkl} \sum_{\mathbf{k}_i, \mathbf{k}_j, \mathbf{k}_a} \hat{\rho}_{b\mathbf{k}_b c}(\mathbf{k}_b + \mathbf{q}_6)(\mathbf{G}) \hat{\rho}_{k(\mathbf{k}_i + \mathbf{q}_6)i\mathbf{k}_i}(-\mathbf{G}) \\ &\quad \times t_{kj}^{ac}(\mathbf{k}_i + \mathbf{q}_6, \mathbf{k}_j, \mathbf{k}_a). \end{aligned} \quad (2.7)$$

With the same practical approximation, the corresponding expressions are

$$\tilde{\mathcal{S}}_2(\mathbf{q}_2 + \mathbf{G}) = \sum_{ijab} \sum_{\mathbf{k}_i, \mathbf{k}_j, \mathbf{k}_a} \hat{\rho}_{a\mathbf{k}_a a}(\mathbf{k}_a + \mathbf{q}_2)(\mathbf{G}) \hat{\rho}_{b\mathbf{k}_b b}(\mathbf{k}_b - \mathbf{q}_2)(-\mathbf{G}) t_{ij}^{ab}(\mathbf{k}_i, \mathbf{k}_j, \mathbf{k}_a), \quad (2.8)$$

$$\tilde{\mathcal{S}}_3(\mathbf{q}_3 + \mathbf{G}) = \sum_{ijab} \sum_{\mathbf{k}_i, \mathbf{k}_j, \mathbf{k}_a} \hat{\rho}_{a\mathbf{k}_a a}(\mathbf{k}_a + \mathbf{q}_3)(\mathbf{G}) \hat{\rho}_{i(\mathbf{k}_i + \mathbf{q}_3)i\mathbf{k}_i}(-\mathbf{G}) t_{ij}^{ab}(\mathbf{k}_i, \mathbf{k}_j, \mathbf{k}_a), \quad (2.9)$$

$$\tilde{\mathcal{S}}_4(\mathbf{q}_4 + \mathbf{G}) = \sum_{ijab} \sum_{\mathbf{k}_i, \mathbf{k}_j, \mathbf{k}_a} \hat{\rho}_{b\mathbf{k}_b b}(\mathbf{k}_b + \mathbf{q}_4)(\mathbf{G}) \hat{\rho}_{j(\mathbf{k}_j + \mathbf{q}_4)j\mathbf{k}_j}(-\mathbf{G}) t_{ji}^{ba}(\mathbf{k}_j, \mathbf{k}_i, \mathbf{k}_b), \quad (2.10)$$

$$\tilde{\mathcal{S}}_5(\mathbf{q}_5 + \mathbf{G}) = \sum_{ijab} \sum_{\mathbf{k}_i, \mathbf{k}_j, \mathbf{k}_a} \hat{\rho}_{a\mathbf{k}_a a}(\mathbf{k}_a + \mathbf{q}_5)(\mathbf{G}) \hat{\rho}_{j(\mathbf{k}_j + \mathbf{q}_5)j\mathbf{k}_j}(-\mathbf{G}) t_{ji}^{ba}(\mathbf{k}_j, \mathbf{k}_i, \mathbf{k}_b), \quad (2.11)$$

$$\tilde{\mathcal{S}}_6(\mathbf{q}_6 + \mathbf{G}) = \sum_{ijab} \sum_{\mathbf{k}_i, \mathbf{k}_j, \mathbf{k}_a} \hat{\rho}_{b\mathbf{k}_b b}(\mathbf{k}_b + \mathbf{q}_6)(\mathbf{G}) \hat{\rho}_{i(\mathbf{k}_i + \mathbf{q}_6)i\mathbf{k}_i}(-\mathbf{G}) t_{ij}^{ab}(\mathbf{k}_i, \mathbf{k}_j, \mathbf{k}_a). \quad (2.12)$$

We will fit to these structure factors instead of the ones in the previous chapter.

Bibliography

- [1] Xin Xing and Lin Lin. Inverse volume scaling of finite-size error in periodic coupled cluster theory. *Physical Review X*, 14(1):011059, March 2024.